

A glossary of definitions of commonly used terms in decision theory, as defined in the current Decision Analysis curriculum.

The task of coming to terms to find a common vocabulary in our field.

Look up "key terms" at the end of each chapter from *Foundations of DA*

I. Math

Event algebra

: Boolean algebra of events. A two valued algebra of True and False. (French, Ch. 6)

Rules of actional thought

: Five rules derived from the axioms of choice

- **probability** you use possibilities and probabilities to provide the distinctions and information that characterize deals.
- **order** You can arrange prospects in a list in descending order of your preference from best to worst.
- **equivalence** you can specify a number p so that you will be indifferent between receiving prospect B for sure and receiving a deal with probability p of prospect A and probability $1-p$ of prospect C where $A > B > C$.
- **substitution** The substitution rule requires that you be indifferent between receiving any prospect and any deal for which you have established that prospect as a certain equivalent.
- **choice**. The rule says that you must choose the deal with the higher probability of the prospect you like better.

Probability "calculus"

Law of total probability

negation $P(A) = 1 - P(\sim A)$

$P(\text{certain event}) = 1$

maximum expected utility

II. Variable

Variables are one of three kinds, an uncertainty, a value (of an outcome), or a decision. The properties each express are distinct. Outcomes and decisions are not uncertain, Uncertainties and values are not under a decision-makers control, and Uncertainties and decisions are "value free" in the sense that all preferences are expressed by values.

clarity

event

: a proposition, and a point in time, whose truth value is determined when it occurs.

sample space, event spaces

: The sample space is the union of all events, equivalent to the certain event.

sample,

possible worlds

mutually exclusive - disjoint partition, exhaustive

characterization (p 62)

: describing the futures for each alternative we face - by creating **possibilities** for each future

possibilities

: possibilities are sometimes referred to as *events* or outcomes (Ch. 5, p 158) and they combine, as with conventional Boolean algebra, with "not", "and" & "or." "An **elemental possibility** (an *atomic state*) is composed of one degree of each kind of distinction (p66)" means that the combination of the events from each uncertain variable generates the full set of all possible events. Consider a path through the probability tree to the terminal node.

possibilities, degrees <-> events, states, outcomes

degrees of distinctions <-> variables, nodes

possibility tree

: A tree with one probability node, for each "distinction"

prospects

: "prospects characterize the future created by the choice of an alternative and any one of its possibilities" Its a possibility (an event) to which a payoff is ascribed (a utility) that is a consequence of both the decision and a subsequent uncertainty.

distinctions (p 60)

: "a division of reality into **possibilities**" (p 62) To form "a precise decision language" (ch 4) See G.S. Brown "Laws of Form" (1969)

: Often used to mean "terms" or "terminology", as in "powerful distinctions." Words with precise meanings. A definition that tells one term from another: "a distinction between"

: Distinctions are named, (age, weight, market share) hence they are variables. A distinction enumerates a set of possibilities (p 62) Mutually events partition the sample A distinction must have **clarity** (be clear), **observability** and **usefulness**. "*Distinction*" is used to mean a **variable**, e.g. such as a node in a probability tree.

clarity test

: A distinction clear enough so that the clairvoyant can tell without exercising judgment whether

the distinction (sic, e.g. *event*) has occurred.

observability

: A "distinction" (event) that is already known, or will be known (observed). For instance, an event observable at the time a decision will be made.

expert, expertise

: "they understand powerful distinctions about the subject of interest" (p 58)

simple distinction

: "only two possibilities" or **two degrees**; a binary state variable (p 65, 123). "Degrees with multiple degrees of distinction," i.e. variables with multiple states

degrees (e.g states) of distinction(p91, 65)

: number of states of a variable, also used to mean the set of states. Like "Distinction", it can mean either an individual state or the r.v.'s set of states. The "degrees of a distinction" must be "**collectively exhaustive** and **mutually exclusive**. That is a *partition*.

measures

(used in the sense of a value, winnings, or payoff for an outcome.) In short, associating a value with each possibility of a distinction.

Outcome

means - ends distinction

(Describe the good decision vs good outcome distinction using a decision table of alternatives v/s outcomes, showing that the best row and the best cell do not necessarily correspond.)

as in decision v/s outcome

prospect

: One path of a choice followed by an uncertainty. A certain outcome of the pair of the decision event followed by the uncertain event.

decision basis

: decisions, values, uncertainties - the "three legged stool", that "represents the *decision frame*"
"the decision frame determines the basis"

III. Decision

"what you can do"

Decision

: "A decision is a choice between two or more alternatives that involves an irrevocable

allocation of resources" (p 8) An act that has tangible consequences. A decision's choices form an "either-or," and once a decision is made, resources are allocated, so reversing a decision would incur some cost.

deliberative vs affective

: Deliberative means "reasoned" Also popularized by Kahneman as 'thinking slow' versus thinking fast

declaring a decision

: Decisions imply intentionality and agency. "We can declare decisions proactively without any external stimulus (P4)" Agency - the decision is under your control.

alternative

: One choice from a set that make up a decision. "Choice" has the sense of a response, after the decision is made, and "alternative" as a possibility, before.

logical reasoning

: Rational choice. Alignment of means and ends. One's actions are taken to achieve one's desired outcomes.

single person, individual decision making

prospect theory

IV. Uncertainty

"what you know"

Uncertainty is a pervasive aspect of any "real world" quantity. Uncertainty is measured by probability, and the terms are often interchangeable.

degree of belief

: "probability is a degree of belief" determined by a "probability assignment"

: The quantification of degree of belief with probability extends also to the more limited concept of frequency as observed ratios. The euphemistic terms "confidence", "likelihood", "noise", "variance", "credibility", "propensity" are often used imprecisely as synonyms.

deal

: A lottery, bet - a gamble over a set of outcomes with specified probabilities. A distribution over a set of prospects, consequent to a choice. "Each alternative produces a deal (p158)"

: A deal is composed of various future prospects. "each alternative produces a deal" (p158)

risk

: in common usage risk is loosely synonymous with uncertainty, but risk often implies both uncertainty and value, typically a possible loss. The term is used as an adjective. e.g. risk

aversion, not as a noun. "By **risk** we mean a prospect that has both a small probability of occurring and a large loss. (p 648)"

probability mass,

"non-deterministic",

statistical independence

probability trees,

discrete distributions

conditional probability

random variable

: A random variable is a *function* (mapping) from the space of "possible worlds" to events to which probabilities can be ascribed.

Information

: "what we know" Lack of information implies uncertainty. (But this explanation does not define information.)

: A fact / distinction that is revealed; The opposite (resolution) of uncertainty. (Knowledge and information are terms used interchangeably.)

Information is one's knowledge encoded in the distinctions one makes, such that a probability distribution can be used to describe it. In short information implies a representation of knowledge as an r.v.

imperfect information

: information that does not yield clairvoyance" (p 30) (This definition is circular.) Use the probabilistic definition of a "garbling" instead,

material Information in a decision problem changes a decision where the information is available. (We need a term for the affected decision.)

Value of clairvoyance

" The PIBP of paying for the clairvoyant's service (p 47)"

The clairvoyant personifies a hypothetical that removes a specified uncertainty. The metaphor of the clairvoyant explains a kind of counterfactual. (S)he knows the future state of a distinction with certainty, and will tell it truthfully if asked. The answer she gives is "information". Such value of information (VOI) is called *the value of perfect information*. (ch 13) This computation is often illustrated by switching the order of decision and uncertainty nodes in a tree. Often it is net of a cost of acquiring (paying the clairvoyant) for the revelation of the information.

For a person with a utility with the "delta property" VOI can be computed as the difference of certain equivalents with and without information.

Probability

: A "unit" measure of uncertainty following "finite additivity", conditioned on state of information.

state of information

data

Mutual information,
cross entropy

V. Value

"what you want"

"**preferences**" that arise from different alternatives. (p16)" Without preferences one would be *indifferent* to outcomes.

bid, bidding (French p 24)

See PIBP

cycle of ownership

: "Within a given cycle of ownership, buying and selling prices must be the same (p 45-6)"

Personal Indifferent Buying Price (PIBP) and Personal Indifferent Selling Price (PISP p.42) "rests on the notion of a fungible, divisible and alienable value measure (p 195)" These are not the same as the market price - the "value in exchange" that is determined when two people transact a trade s.t. one's PIBP is equal or greater than another's PISP. (? Are these "reservation prices"?)

The wealth effect when the delta property doesn't hold breaks the equivalence between PIBP and PISP.

The instantaneous equivalence of PISP and PIBP as expressed by "cycle of ownership" does not necessarily hold if time, information or wealth differ between the two exchanges.

(Is the wealth effect captured entirely by the risk preference of the utility function or is there a marginal substitution for wealth for different goods that explains it?)

value in use versus value in exchange

: (p 50)

sunk cost principle

: e.g. what you sell something for doesn't depend on what you paid for it. (p.28)

Utility

preference probability

: Probability assigned to a certain outcome (a prospect). A consequence of the equivalence rule

- the utility of a prospect. There is both a "preference probability of a prospect" -- a utility, and the "preference probability of an alternative" -- an e-value of the prospects' utilities. Preference is quantified by a utility function, which can express time and risk preference to the value of an outcome. Preference probabilities are a consequence of the *equivalence rule*.

u-curve

: A **Utility function** that maps e.g. dollars into preference probabilities (utilities on a scale set by the min and max prospects used in elicitation.)

e-value (equivalent value)

: a probability weighted sum of preference probabilities, the consequence of an "e-operation" -- an **expected value**, by taking an **expectation**. Also in probability terms the *mean value* of the distribution. The "e-value" of a deal with a u-curve is a **u-value**, e.g. the "e-value of a u-value."

delta property

: a Utility function that has constant risk aversion for changes in wealth.

risk odds

: Given an exponential utility function, the ratio $r_m = \frac{p}{1-p}$ where p is the preference probability for a reward of m , versus a loss of m . r is the base to the exponent in the utility function $u(x) = a - br^{-x}$. When risk neutral, this ratio is equal to 1 and the function reduces to $u(x) = a + bx$.

- risk aversion coefficient γ (p 231)
- risk tolerance coefficient ρ

certain equivalent

: the PISP, possibly for an uncertain deal. Computed by taking the inverse of the utility function.

time preference

VI. Dependence

relevance (p 123)

: when knowledge of one distinction (rv) changes belief in another distinction (rv)

the mutual property (p 125)

: that relevance is non-directional ("mutual"), unlike cause that is directional. This is called the "associative vs distinctive logic"

Cause

: The means by which we understand change. An explanation is just a description of a cause. This is a case where the decision analysis community invented euphemisms -- "influence",

"relevance" -- to avoid using the obvious term. Precisely a cause is defined by a hypothetical -- knowing what would be the difference in the consequence should the event in question not have occurred -- the "counterfactual." Resolving whether a cause exists or not is the point of scientific method, in those cases where cause is not apparent.

Likelihood ratios

multivariate, joint distributions, correlation

hypotheticals / counterfactuals.

Model

Markov networks,

Bayes networks,

causal graphs,

structural equation models

dynamic programming

VII. Quality, axioms

clarity of action

: clarity of action (p 13) implies that the best decision possible is made. It is assured by using correct methods ("instruments for decision-making.") Having clarity of action removes *regret*.

formulation

frame

: "A way of viewing the decision (p15)". Borrowing from systems engineering, the boundary of the system is chosen to include what is relevant to modelling the decision, and leaving out what is unnecessary

bias

Decision Quality

Main elements of a **good decision**: (p 15)

- Decision-maker
- Frame
- Alternatives
- Preferences
- Information
- Logic

Ethics

normative vs descriptive theory

: A descriptive theory is predictive, specifically, able to predict people's behaviors. A normative theory says what to do to meet one's goals. "principled" implying it follows from a axioms of choice

Ethics

: In popular terms, implies one's duties and behavior toward a group -- to society at large.

: The terms ethics and morals are often used interchangeably, sometimes with the distinction of "personal morality" and "social ethics"

social norms